

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Alpha-911 MLOps Governance & Observability Layer	
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Abstract

Problem Statement: AI-driven trading signal systems evolve rapidly — thresholds get retuned, confluence rules get adjusted, and models are redeployed frequently in response to changing market conditions. Without a formal MLOps layer, this evolution happens ad hoc: there is no versioned record of which configuration produced which real-world results, no automated way to detect when a signal engine's performance is silently drifting, and no structured check for whether the model performs unevenly across different market regimes. This makes it difficult to trust, audit, or safely iterate on a live signal engine over time.

This project builds an MLOps governance and observability layer around an existing AI trading signal platform (Alpha-911) that generates signals for crypto and NSE/BSE markets using CPR/pivot confluence, macro kill-switch logic, and OI/funding context. DVC is used to version each iteration of the signal engine's configuration and parameters, so every change is tracked and reversible. MLflow logs each deployed version alongside its paper-trading performance, creating an auditable history of what was deployed, when, and how it performed. A Prometheus and Grafana pipeline continuously monitors live signal accuracy and win-rate, flagging performance drift before it becomes a costly problem. Finally, a fairness-analysis module evaluates whether the signal engine performs consistently across different conditions — such as high versus low volatility periods, and crypto versus NSE markets — surfacing any systematic blind spots. Together, these components turn an already-functioning AI trading system into a properly governed, observable, and auditable MLOps pipeline, aligning a real production system with responsible AI deployment practices.

Signature of the Guide

HOD